
EE1060: Differential Equations and Transform Techniques

Lecture 22: Fourier Series

Topics :

1. Symmetry and its impact on Fourier Coefficients
 2. Effect of time-shift and Derivative on Fourier Coefficients
-

Fourier Coefficients for Even and Odd functions

① If $f(x)$ is real and even, the complex Fourier coefficients are purely real.

Proof :- a) Since $f(x)$ is real

$$C_{-k} = C_k^* \rightarrow \textcircled{1}$$

from ① & ②

b) Since $f(x)$ is even

$$[C_{-k} = C_k] \rightarrow \textcircled{2}$$

$$C_k = C_k^* \Rightarrow C_k = \text{Real.}$$

$$C_{-k} = \frac{1}{L} \int_0^L f(x) e^{+jk\omega_0 x} dx \leftarrow \text{Analysis Eqn}$$

using Synthesis Eqn:-

$$f(x) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 x} \rightarrow \textcircled{3}$$

$$f(-x) = \sum_{n=-\infty}^{\infty} C_n e^{-jn\omega_0 x} \rightarrow \textcircled{4}$$

$\therefore f(x)$ is even

$$f(x) = \frac{1}{2} [f(x) + f(-x)]$$

$$\textcircled{3} = \frac{1}{2} [\textcircled{3} + \textcircled{4}]$$

$$= \frac{1}{L} \int_0^L f(-u) e^{-jk\omega_0 u} (-du) \quad \text{Sub } x = -u$$

$$= \frac{1}{L} \int_{-L}^0 f(u) e^{-jk\omega_0 u} du \quad \text{Sub } v = u + L.$$

$$= \frac{1}{L} \int_0^L \underbrace{f(u+L)}_{f(L)} e^{-jk\omega_0 v} \underbrace{e^{jk\omega_0 L}}_{\textcircled{1}} dv$$

$$= C_k$$

Effect of time shift on Fourier Coefficients

given: signal $f(t) \rightarrow \{c_k\}_{k=-\infty}^{\infty}$ Set of Fourier Coefficients $f(t) = a_0 + \sum [a_n \cos n\omega_0 t + b_n \sin n\omega_0 t]$

$f(t-\tau) \rightarrow \{c'_k\}$ $\tau > 0 \rightarrow$ moving the signal to right
 $\tau < 0 \rightarrow$ " " " " left

$$c'_k = \frac{1}{L} \int_0^L f(t-\tau) e^{-jk\omega_0 t} dt$$

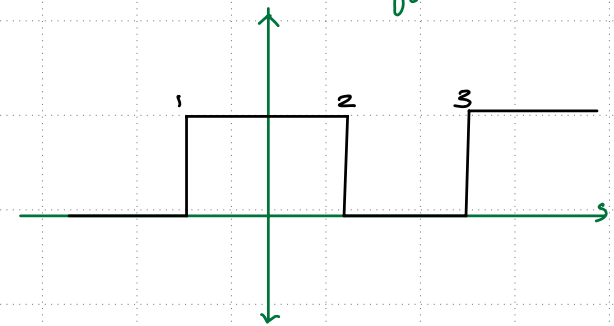
Sub $u = t - \tau \Rightarrow$

$$c'_k = \frac{1}{L} \int_{-\tau}^{L-\tau} f(u) e^{-jk\omega_0 u} \underbrace{e^{-jk\omega_0 \tau}}_{\text{constant}} du$$

$$= e^{-jk\omega_0 \tau} \underbrace{\frac{1}{L} \int_{-\tau}^{L-\tau} f(u) e^{-jk\omega_0 u} du}_{c_k}$$

$$c'_k = e^{-jk\omega_0 \tau} c_k$$

$$|c'_k| = |c_k| \quad \& \quad \angle c'_k = \angle c_k - k\omega_0 \tau$$



Effect of Derivative on Fourier Coefficients

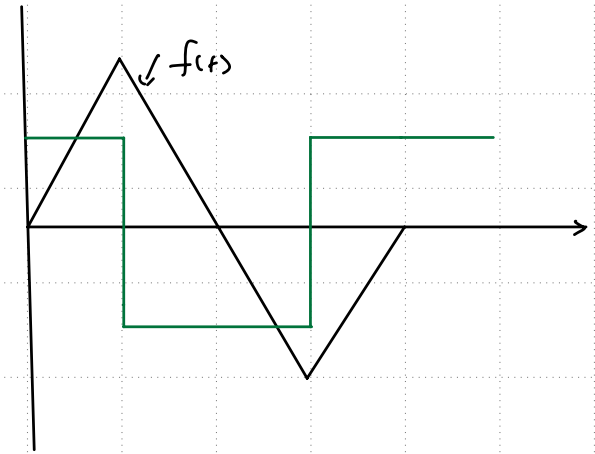
given: signal $f(t) \rightarrow \{C_k\}_{k=-\infty}^{\infty}$ Set of Fourier Coefficients

$f'(t) \rightarrow ?? C'_k$

$$f(t) = \sum_{n=-\infty}^{\infty} C_k e^{jk\omega_0 t}$$

$$\underbrace{f'(t)} = \sum_{n=-\infty}^{\infty} \underbrace{C_k (jk\omega_0)}_{C'_k} \underbrace{e^{jk\omega_0 t}}$$

$$C'_k = (jk\omega_0) C_k.$$



$$\left[\begin{array}{l} \text{for Sawtooth} \\ \text{wave} \end{array} \right] C_k = \frac{A}{\pi k} (e^{j\pi k} - 1)$$

Convergence of Fourier Series

Pointwise Convergence:-

Original signal $f(t)$

reconstructed signal $S_N(t) = \sum_{n=-N}^N c_n e^{jn\omega_0 t}$

if as $N \rightarrow \infty$ $S_N(t) \rightarrow f(t)$ at every point.

then FS Converges in point-wise sense.

3 Conditions:- $f(t)$ is periodic

a) $f(t)$ is bounded

b) $f(t)$ must have finite no. of maxima & minima

c) $f(t)$ must have finite no. of discontinuities

if (3) Cond hold, then

$$S_N(a) = f(\underline{a}) \quad \text{if } f(t) \text{ is continuous at } a$$

$$S_N(a) = \frac{f(a-) + f(a+)}{2} \quad \text{if } f(t) \text{ is discontinuous at } a$$

Left hand limit Right hand limit

N for various points $a \in I$

↳ indicator for rate of convergence.